BEGINNER COURSE 2025

PYTHON FOR DATA SCIENCE

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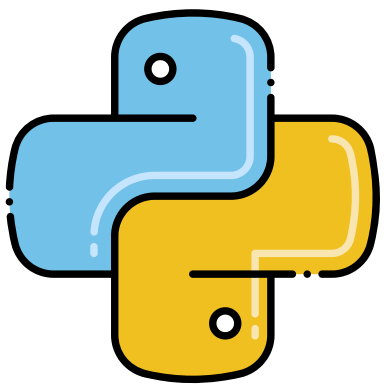
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Advantages

- Instantly convert between shapes (1D → 2D → 3D)
- Essential for ML pipelines, image and signal processing
- Enables batch operations for models
- Cleaner, vectorized code = fewer bugs
- Makes complex operations feel intuitive

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Example

```
"""
Example: Shaping/Resizing
         on Numpy Arrays
"""

import numpy as np

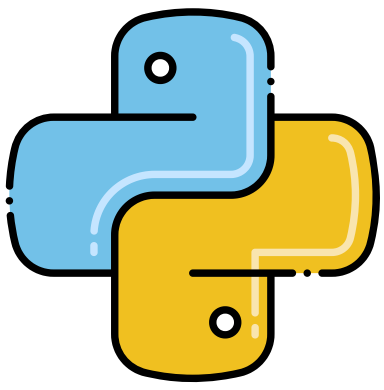
a = np.arange(12)
print(a.shape)  # (12,)

b = a.reshape(3, 4)
print(b)
# Output: 3x4 array

c = b.flatten()
print(c.shape)  # (12,)
```

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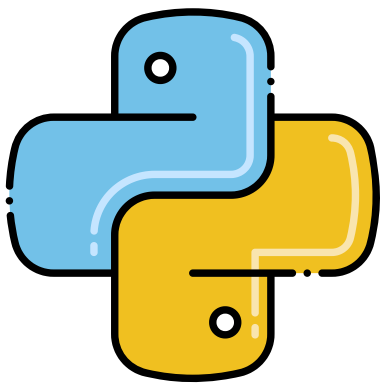
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Why is this powerful?

- Memory efficient: Views instead of copies
- Time-saving: Shape your data in milliseconds
- Model-ready: Feed your neural nets without preprocessing nightmares
- Versatile: Works seamlessly with pandas, TensorFlow, and beyond.

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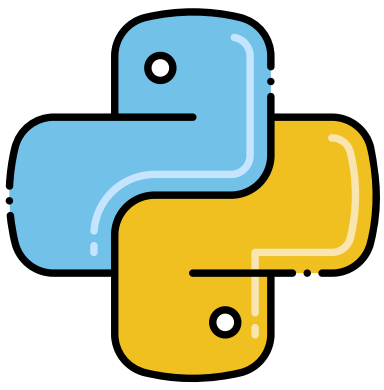


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Once I mastered reshaping in NumPy, data manipulation became second nature. I could prototype ideas faster, with cleaner, smarter code.

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Fun Fact

- reshape() doesn't just transform arrays — it transforms how you think about dimensions.

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👊 Ready to level up your NumPy skills?

👇 Comment your favorite trick or challenge with reshaping arrays!

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